Physics-Informed Machine-Learning Inversion of Chandrayaan-3 RAMBHA-LP Langmuir Probe Sweeps

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The RAMBHA-LP Langmuir probe on the Chandrayaan-3 Vikram lander recorded current–voltage sweeps of the near-surface lunar plasma near the south pole between 24 August and 3 September 2023. Electron density, electron temperature and plasma potential are conventionally extracted by piecewise curve fitting, which fails on 44% of the 29,862 sweeps in the public rawB archive. We trained a physics-informed neural network on 40,000 synthetic sweeps from an orbital-motion-limited forward model, with noise and parameter ranges measured from the real archive, and a loss that also penalises failure to reconstruct the input curve. 30% of the sweeps are clipped by the instrument and neither method can read them, so they are excluded; the network recovers 5,115 of the 5,641 unclipped classical failures (91%), raising usable coverage from 50.7% to 67.9%. With no ground truth on the Moon, we validate against an independent classical fit (no meaningful density offset; 4.4% per sweep over 15,147 unclipped sweeps), against the classical value on the other half of the same sweep for 2,157 recovered sweeps (5.3%), and by self-consistency between the rising and falling halves of each commanded sweep (2.4%, r = 0.992, 9,895 pairs); the daily medians also follow the mission-long density rise from 245 to 500 cm^{-3}, as the classical fit does. Median values agree with the published ranges for this instrument, except for density on the first three days. Comparing the six channel and resistor setups, which sample the same plasma seconds apart, shows two reading density low, by about 17% and, early in the mission, by about half; both methods share these offsets. Physics-informed training worsens accuracy on synthetic data, yet on real sweeps it makes the two halves of each sweep 30% more consistent and removes a 5% density offset against the classical fit, showing that simulation alone cannot judge sim-to-real choices. A two-population extension with a working negative control finds no evidence for a second electron population, in agreement with the instrument team, and shows that the standard averaged-sweep diagnostic for multiple populations is not valid evidence on its own.